CNENGR202 Instrumentation and Control

Part 1 L4 – Laplace Transforms, Transfer Functions

Prof. Zhang Liwei

Email: lwzhang@bjtu.edu.cn
Working Room: SX702

School of Electronics and Information Engineering
Beijing Jiaotong University

Outline

- **Introduction**

- **Laplace Transforms**
  - Worked Example (MSD)
  - Worked Example (RLC)
  - Transfer Functions

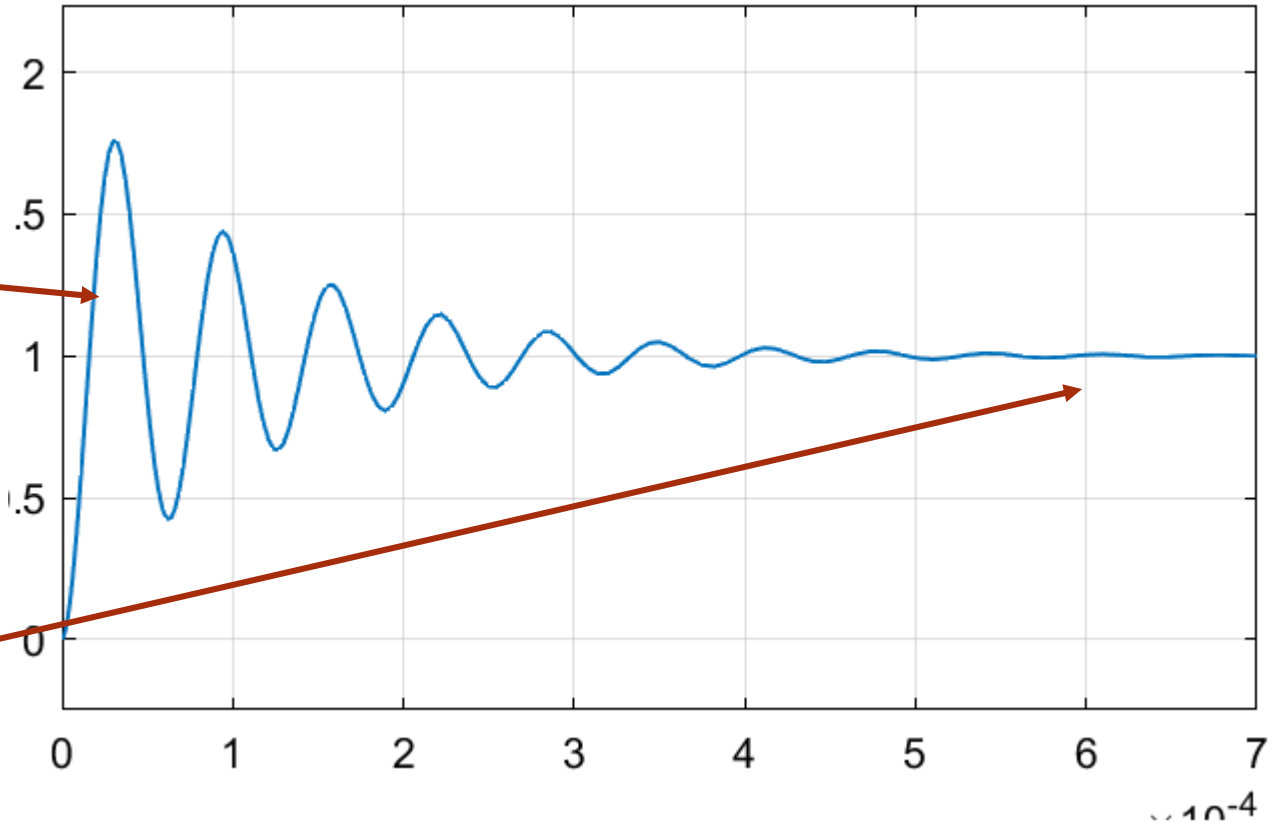
- **Stability Analysis**
  - Poles & Zeros

Introduction - 1

- The graph shows the 3rd order response of a wind turbine model to a unit step input. The input and output variables are the blade angle and torque respectively

Second order oscillatory mode

First order very slow exponential decay



Introduction - 2

- Systems can often be represented using 1st or 2nd order models
- However, higher order equations are sometimes necessary to describe more complex dynamic behaviour, so we should learn about the solution to generalized differential equations
- Learning Objectives. At the end of this section, you should be able to:
  - Use Laplace to solve simple differential equations; use the Final Value Theorem; and write down the Transfer Function
  - Determine the poles and zeros of a Transfer Function; and understand the dominant response modes
  - Use the rules of block diagram manipulation

Introduction - 3

- Consider the 1st order model

$$\tau \frac{dx}{dt} + x = Ku(t) \qquad a_1 \frac{dx}{dt} + a_0 x = b_0 u$$

- Similarly, for a 2nd order system

$$\frac{d^2x}{dt^2} + 2\zeta\omega_n \frac{dx}{dt} + \omega_n^2 x = Ku(t) \qquad a_2 \frac{d^2x}{dt^2} + a_1 \frac{dx}{dt} + a_0 x = b_0 u$$

Introduction - 4

- Here is a 3rd order system

$$a_3 \frac{d^3x}{dt^3} + a_2 \frac{d^2x}{dt^2} + a_1 \frac{dx}{dt} + a_0 x = b_0 u$$

- Sometimes we need to consider rates of change of the input

$$a_2 \frac{d^2x}{dt^2} + a_1 \frac{dx}{dt} + a_0 x = b_1 \frac{du}{dt} + b_0 u$$

Introduction - 5

- Generalised Linear Differential Equation

$$a_n \frac{d^n x}{dt^n} + a_{n-1} \frac{d^{n-1} x}{dt^{n-1}} + \cdots + a_1 \frac{dx}{dt} + a_0 x$$
$$= b_m \frac{d^m u}{dt^m} + b_{m-1} \frac{d^{m-1} u}{dt^{m-1}} + \cdots + b_1 \frac{du}{dt} + b_0 u$$

- This is an *n*-th order system (the terminology refers to the output)

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Laplace Transforms - 1

- Overview of **Laplace Transforms**

$$L\left[f(t) \right] = F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

$$L^{-1}\left[F(s) \right] = f(t) = \frac{1}{j2\pi} \int_{\sigma - j\omega}^{\sigma + j\omega} F(s) e^{st} ds$$

Properties of Laplace Transforms - 1

$$L\left[af_1(t)+bf_2(t)\right]=aL\left[f_1(t)\right]+bL\left[f_2(t)\right]$$

Superposition

$$L\left[\frac{df}{dt}\right]=sF(s)-f(0)$$

In general $f(0)=0$

Such as current in inductor and voltage in capacitor

$$L\left[\int f(t)dt\right]=s^{-1}F(s)$$

$$L\left[\frac{d^2f}{dt^2}\right]=s^2F(s)-sf(0)-\left(\frac{df}{dt}\right)_{t=0}$$

Properties of Laplace Transforms - 2

- Final Value Theorem

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

(Steady State Value)

- Initial Value Theorem

$$\lim_{t\to 0} f(t) = \lim_{s\to\infty} sF(s)$$

Final Value: Example

$$F(s) = \frac{1}{s(s+5)}$$

$$F(s) = \frac{s+10}{s^3+5s^2+6s}$$

$$F(s) = \frac{2}{s} + \frac{1}{s+2} + \frac{3}{s+3}$$

Initial Value: Example

$$F(s) = \frac{2s + 10}{s^2 + 2s}$$

$$F(s) = \frac{5s^2 + 10}{s^3 + 3s^2 + 2s}$$

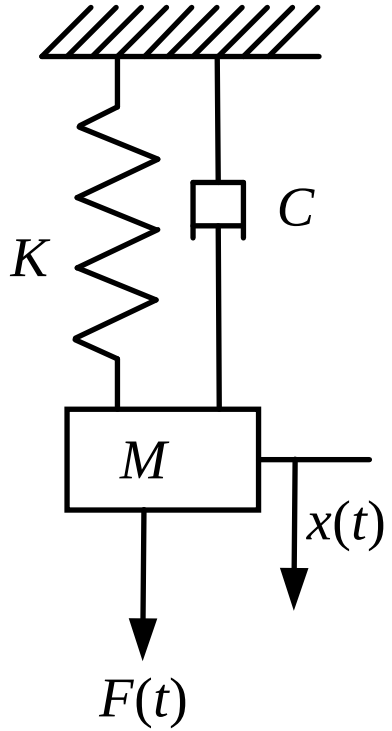
$$F(s) = \frac{1}{s} + \frac{2}{s+2} + \frac{3}{s+3}$$

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Worked example: Mass–Spring–Damper (MSD) - 1

$$M\frac{d^2x}{dt^2}+C\frac{dx}{dt}+Kx=F(t)$$



Worked example: Mass–Spring–Damper (MSD) - 2

$$M\frac{d^2x}{dt^2}+C\frac{dx}{dt}+Kx=F(t)$$

$$M\left(s^2X(s)-sx(0)-\frac{dx(0)}{dt}\right)+C\left(sX(s)-x(0)\right)+KX(s)=F(s)$$

Worked example: Mass–Spring–Damper (MSD) - 3

$$M\left(s^2X(s)-sx(0)-\frac{dx(0)}{dt}\right)+C\left(sX(s)-x(0)\right)+KX(s)=F(s)$$

- To find the Transfer Function, set the initial conditions to zero, i.e.

$$x(0)=0 \qquad \frac{dx(0)}{dt}=0$$

$$Ms^2X(s)+CsX(s)+KX(s)=F(s) \qquad X(s)=\frac{1}{Ms^2+Cs+K}F(s)$$

MSD response to a unit step Input - 1

- From the table, a unit step input is defined

$$U(s) = \frac{1}{s}$$

$$X(s) = \frac{1}{\left(Ms^2 + Cs + K\right)s}$$

- Numerical example

$$M = 1$$
$$C = 3$$
$$K = 2$$

$$X(s) = \frac{1}{\left(s^2 + 3s + 2\right)s}$$

MSD response to a unit step Input - 2

$$X(s) = \frac{1}{\left(s^2 + 3s + 2\right)s}$$

$$X(s) = \frac{1}{s(s+1)(s+2)}$$

$$X(s) = \frac{1}{2s} - \frac{1}{s+1} + \frac{1}{2(s+2)}$$

MSD response to a unit step Input - 3

$$X(s) = \frac{1}{2s} - \frac{1}{s+1} + \frac{1}{2(s+2)}$$

- Take inverse Laplace Transforms

$$x(t) = \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}$$

Determine the steady state value ?

Determine the initial value ?

MSD response to a unit step Input - 4

- The exponentials decay to zero, hence at steady state $x(t \to \infty) = 0.5$

$$x(t \to \infty) = 0.5$$

$$x(\infty) = \lim_{s \to 0} sX(s) = 0.5$$

Final Value Theorem

$$\lim_{x\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

- Continue MSD example

$$\lim_{s\to 0} sF(s) = \lim_{s\to 0}\left(s\frac{1}{(s^2+3s+2)s}\right) = \frac{1}{2} = 0.5$$

This ‘s’ comes from the Final Value Theorem above

Within the Transfer Function set s=0

Laplace transform of a unit step

MSD response to a unit step Input - 5

- The exponentials decay to 1, hence at initial $x(t \to 0) = 0$

$$x(t \to 0) = 0$$

$$x(0) = \lim_{s \to \infty} sX(s) = \lim_{s \to \infty} s \frac{1}{(s^2 + 3s + 2)s} = 0$$

Steady state gain

Alternatively find the steady state gain of the transfer function by setting s=0 and then multiply by the input

$$\lim_{t\to\infty} x\left(t\to\infty\right) = 0.5\times u_0$$

For the case of a unit step u=1

This is the steady state value of the output variable

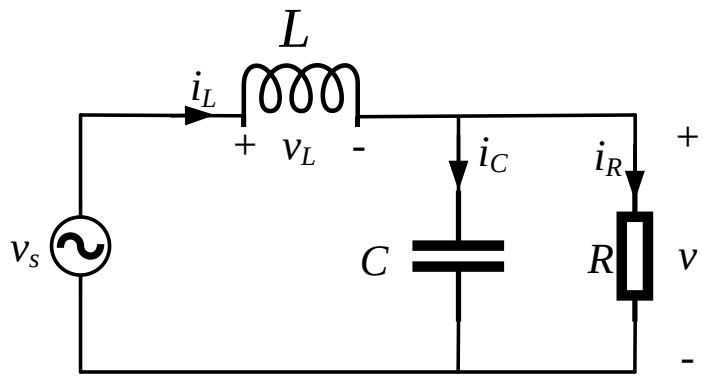
This 0.5 is the steady state gain

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RLC example - 1

$$LC\frac{d^2v}{dt^2}+\frac{L}{R}\frac{dv}{dt}+v=-v_s$$



- if
  - v_s is a unit step signal
  - L=500mH
  - C=100μF
  - R=10Ω

$V(s)$=?
Steady state value?
Initial value?

RLC example - 2

$$V(s) = \frac{\mathbf{20000}}{s\left(s^2 + \mathbf{10}s + \mathbf{20000}\right)}$$

$$v(\infty) = \lim_{s \to \mathbf{0}} s \cdot \frac{\mathbf{20000}}{s\left(s^2 + \mathbf{10}s + \mathbf{20000}\right)} = \mathbf{1}$$

$$v(\mathbf{0}) = \lim_{s \to \infty} s \cdot \frac{\mathbf{20000}}{s\left(s^2 + \mathbf{10}s + \mathbf{20000}\right)} = \mathbf{0}$$

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Transfer function

- Transfer Functions are based on Laplace Transforms with zero initial conditions

$$\frac{dx}{dt} \rightarrow sX \qquad \frac{d^2x}{dt^2} \rightarrow s^2X \qquad \frac{d^nx}{dt^n} \rightarrow s^nX$$

Mass–Spring–Damper System (Recap)

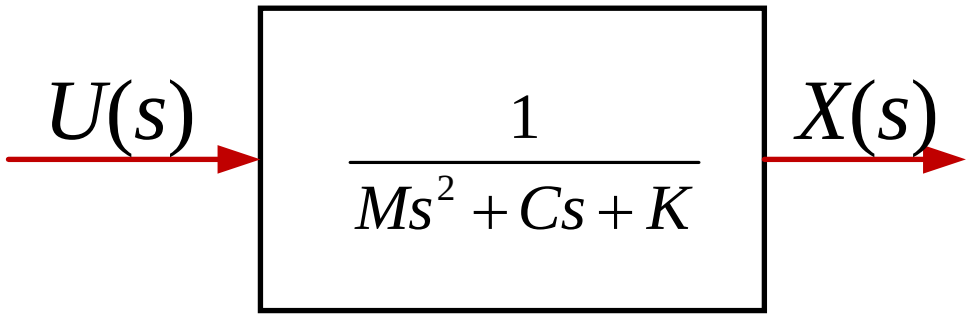
$$M\frac{d^2x}{dt^2}+C\frac{dx}{dt}+Kx=u(t)$$

$$Ms^2X+CsX+KX=U$$

Take Laplace with zero initial conditions

$$X=\frac{1}{Ms^2+Cs+K}U$$

Block diagram form



Determine the TF for this 3rd order system

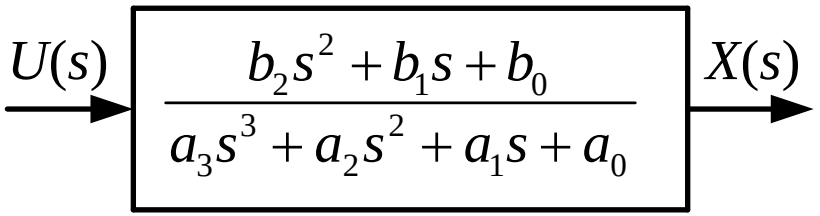
$$a_3\frac{d^3x}{dt^3}+a_2\frac{d^2x}{dt^2}+a_1\frac{dx}{dt}+a_0x=b_2\frac{d^2u}{dt^2}+b_1\frac{du}{dt}+b_0u(t)$$

$$\left(a_3s^3+a_2s^2+a_1s+a_0\right)X=\left(b_2s^2+b_1s+b_0\right)U$$

Take Laplace with zero initial conditions

$$X=\frac{b_2s^2+b_1s+b_0}{a_3s^3+a_2s^2+a_1s+a_0}U$$

Block diagram form



Determine the Steady State Gain

$$X = \frac{b_2 s^2 + b_1 s + b_0}{a_3 s^3 + a_2 s^2 + a_1 s + a_0} U$$

U is a unit step

$$x(t \to \infty) = \lim_{s \to 0} \left(s \frac{b_2 s^2 + b_1 s + b_0}{a_3 s^3 + a_2 s^2 + a_1 s + a_0} \frac{1}{s} \right)$$

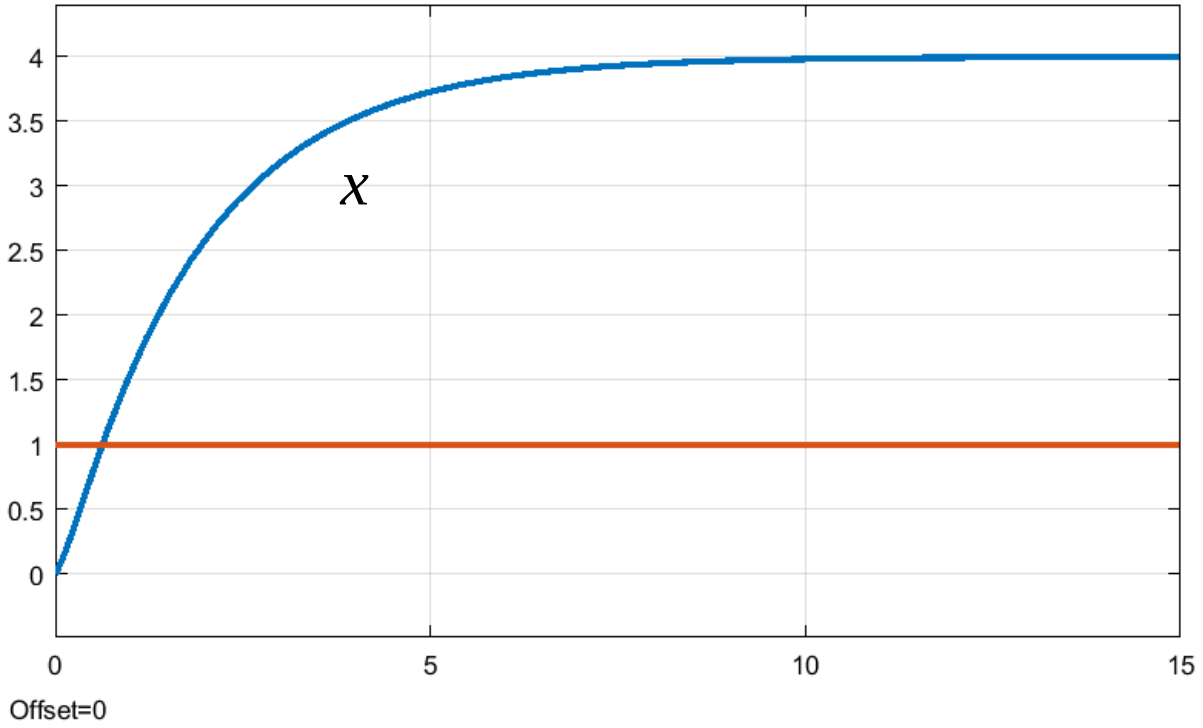
- Determine the steady state gain $\frac{b_0}{a_0}$

Example for 3rd order system

$$X(s) = \frac{s+12}{s^3+6s+3}U(s)$$

If $u(t)$ is a unit step signal

$$x(t \to \infty) = 4$$

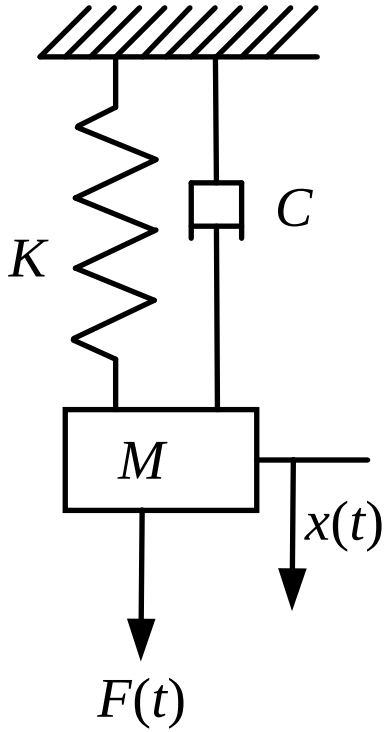


The steady state gain is 4 (set $s = 0$)

MSD example -1

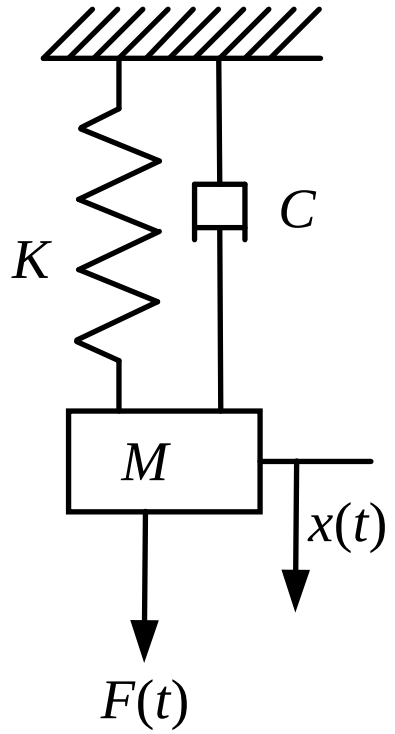
- The displacement *x* represents the distance in metres from an arbitrary reference point.
- If the steady state gain of the mass– spring–damper system is 0.01 and a constant external force of 8N is applied, the inherent frequency is 1, what is the displacement *x* at steady state?

$$\frac{d^2x}{dt^2} + 2\zeta\omega_n \frac{dx}{dt} + \omega_n^2 x = Ku(t)$$



Answer for MSD example

- The displacement *x* represents the distance in metres from an arbitrary reference point.
- If the steady state gain of the mass– spring–damper system is 0.01 and a constant external force of 8N is applied , the inherent frequency is 1, what is the displacement *x* at steady state?



$$\frac{d^2x}{dt^2} + 2\zeta\omega_n \frac{dx}{dt} + \omega_n^2 x = Ku(t)$$

K=0.01

U=8

$\omega_n = 1$

$$x(t \to \infty) = \frac{0.01 \times 8}{1} = 0.08m$$

Outline

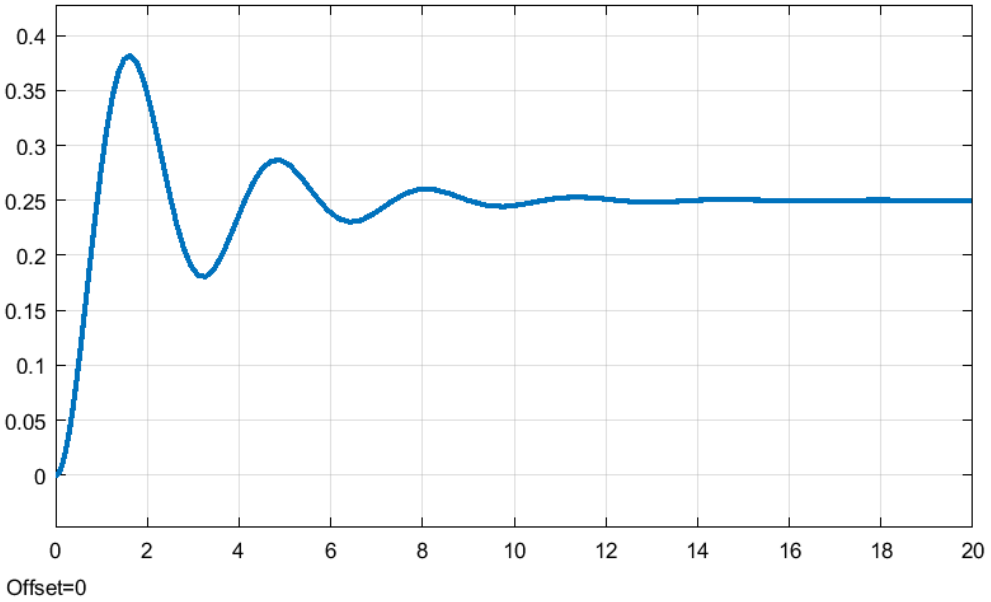
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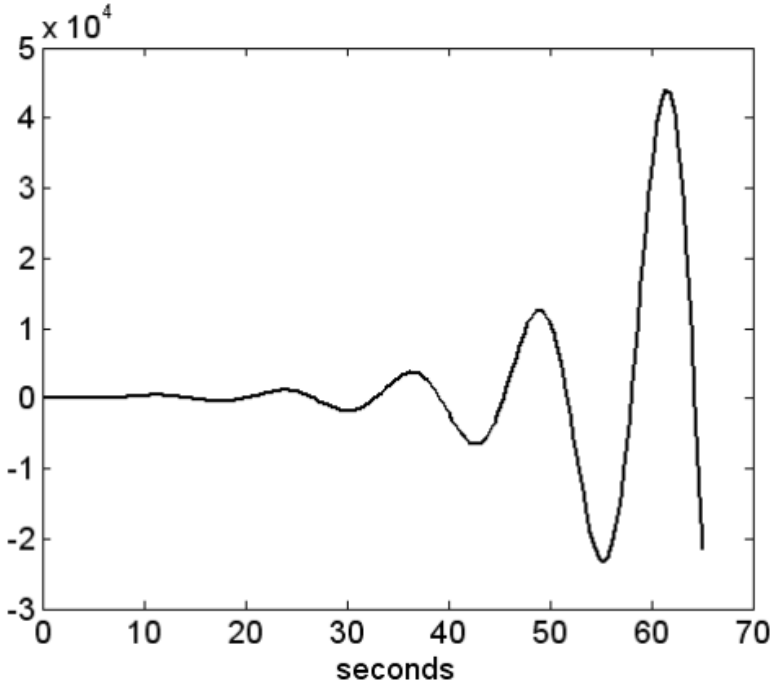
Stable

- One of the most important tasks of a control engineer is to ensure that the closed- loop control system is stable

Stable and unstable

- Definition: “If a system that is initially in equilibrium is disturbed, it is said to be stable if it eventually returns to that equilibrium condition”
  - Upper plot: the response of a poorly damped, oscillatory but nonetheless stable system
  - Lower plot: the response of an oscillatory and unstable system





Asymptotic Stability of a 1st Order Transfer Function

$$\tau\frac{dx}{dt}+x=Ku(t) \qquad (\tau s+1)X=KU \qquad X=\frac{K}{\tau s+1}U$$

- The characteristic polynomial is $\tau s+1$
- Setting this to zero yields the characteristic equation $\tau s+1=0$
- The root (solution) of this equation is called the pole $s=-1/\tau$
- As we have just seen, for stability this pole should be a negative real number i.e. equivalent to a positive τ

This result also applies to higher order systems (with multiple poles)

Example for 1st Order Transfer Function - 1

$$X = \frac{K}{\tau s + 1} U$$

If

K=2

τ=5

$x(t)$=?

Example for 1st Order Transfer Function - 2

$$X = \frac{K}{\tau s + 1} U \longrightarrow X = \frac{2}{5s + 1} U = \frac{0.4}{s + 0.2} U$$

$$\downarrow$$

$$x(t) = 0.4e^{-0.2t}$$

Asymptotic Stability of a Generalised Differential Equation

- Homogeneous equation

$$a_n \frac{d^n x}{dt^n} + a_{n-1} \frac{d^{n-1} x}{dt^{n-1}} + \cdots + a_1 \frac{dx}{dt} + a_0 x = 0$$

- The roots (i.e. solutions of the auxiliary equation) determine the stability

$$a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0 = 0$$

为了使瞬态消失，指数项必须全部减少，即辅助方程的所有解(**根**)必须具有负实部……可以说，分析这一点的最佳方法是使用传递函数模型

- For the transients to die away, the exponential terms must all decrease, i.e. all the solutions (roots) of the auxiliary equation must have negative real parts … arguably, the best way to analyse this is by using the Transfer Function model

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Poles & Zeros – 1

$$a_n \frac{d^n x}{dt^n} + a_{n-1} \frac{d^{n-1} x}{dt^{n-1}} + \cdots + a_1 \frac{dx}{dt} + a_0 x = b_m \frac{d^m u}{dt^m} + b_{m-1} \frac{d^{m-1} u}{dt^{m-1}} + \cdots + b_1 \frac{du}{dt} + b_0 u$$

$$\left(a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0\right) X = \left(b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0\right) U$$

$$X = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0} U$$

$$X = K \frac{\left(s - z_1\right)\left(s - z_2\right) \cdots \left(s - z_m\right)}{\left(s - p_1\right)\left(s - p_2\right) \cdots \left(s - p_n\right)} U$$

Poles & Zeros – 2

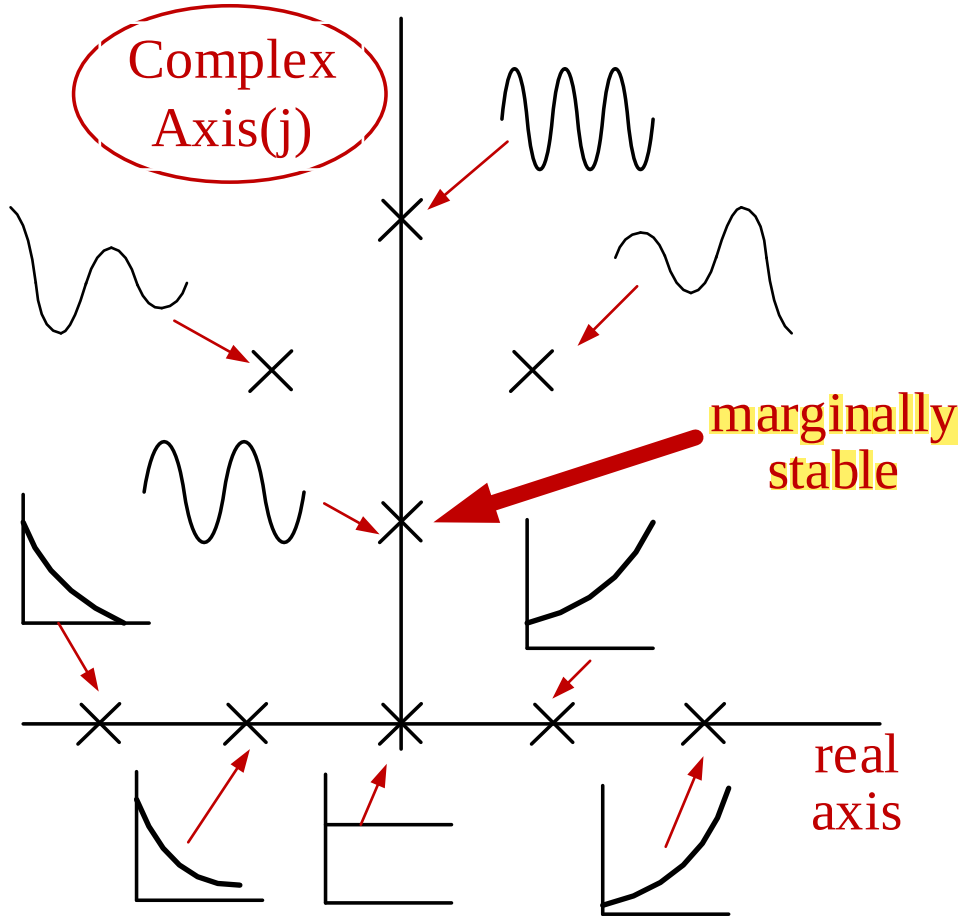
- K is the gain
- $z_{1\ldots}z_m$ are the zeros
- $p_{1\ldots}p_m$ are the poles
- If setting the denominator polynomial to zero, then obtain the characteristic equation

$$(s-p_1)(s-p_2)\cdots(s-p_n)=0$$

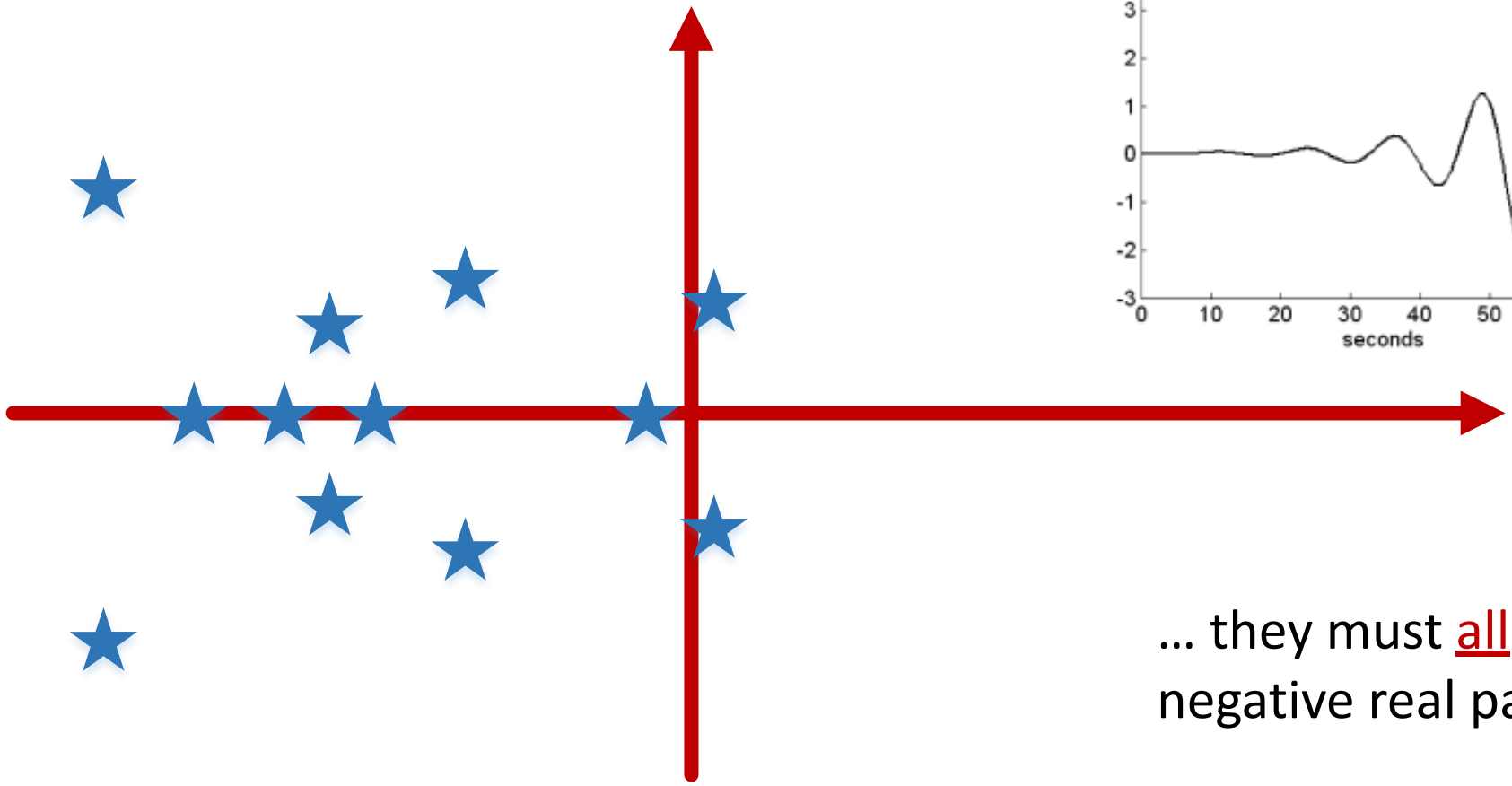
- The poles $p_1, p_2, \ldots, p_n$ are the roots of the characteristic equation
- In general, the poles are complex numbers and, for stability, **they must have negative real parts**

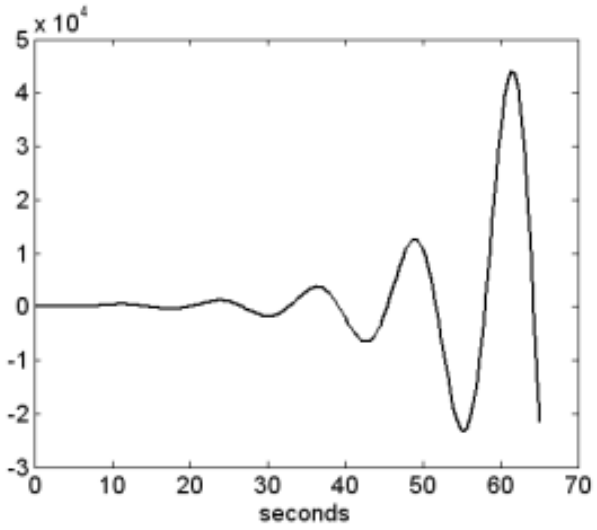
Poles & Zeros – 3

Pole locations and time response modes



Poles & Zeros – 4



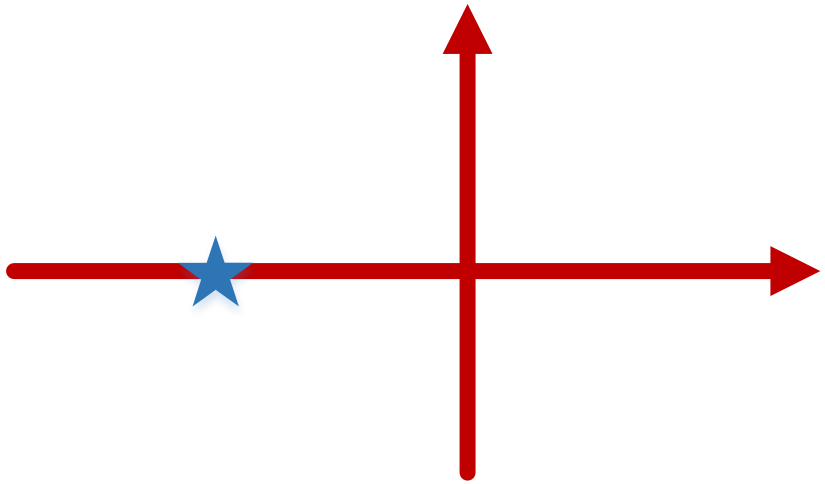


… they must <u>all</u> have negative real parts

First order example

$$10\frac{dx}{dt} + x = u \qquad 10sX + X = U$$

$$X = \frac{1}{10s+1}U \qquad 10s+1=0$$



- Stable

$$s = -\frac{1}{10} = -0.1$$

Second Order Example - 1

$$\frac{d^2x}{dt^2} - \frac{dx}{dt} + 0.5x = \frac{du}{dt} + 2u$$

- Find the **poles** and **zeros**, plot these on the complex *s*-plane, determine the stability and sketch the response

Second Order Example - 2

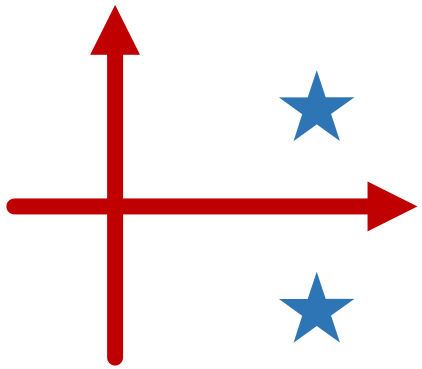
- Take Laplace with zero initial conditions $s^2X - sX + 0.5X = sU + 2U$

- Zeros $s + 2 = 0$ $s = -2$

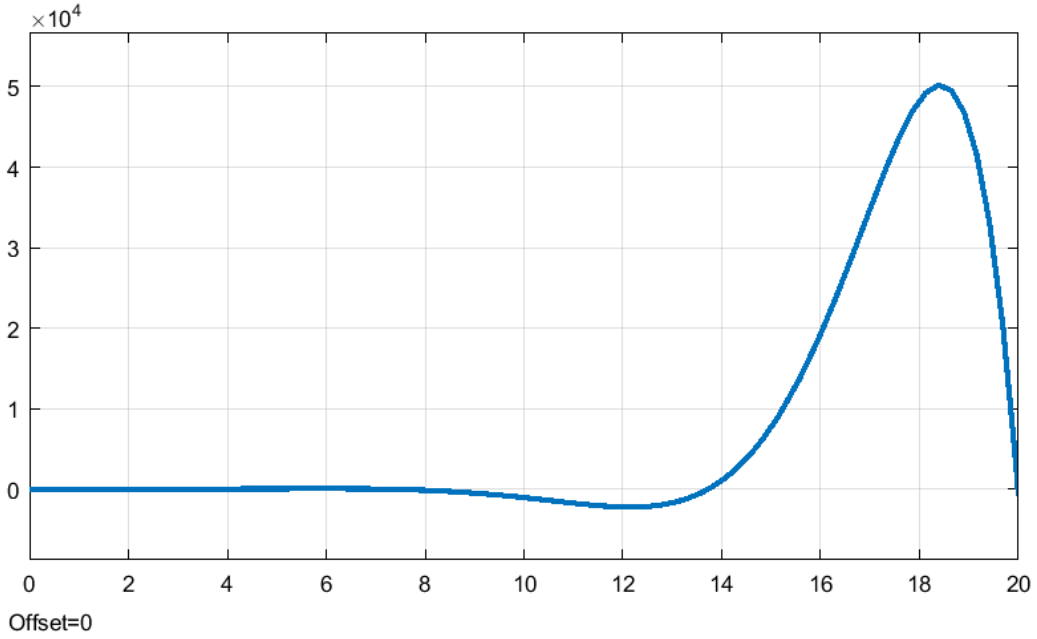
$$X = \frac{s+2}{s^2 - s + 0.5}U$$

- Poles $s^2 - s + 0.5 = 0$

$$s = 0.5 \pm 0.5j$$

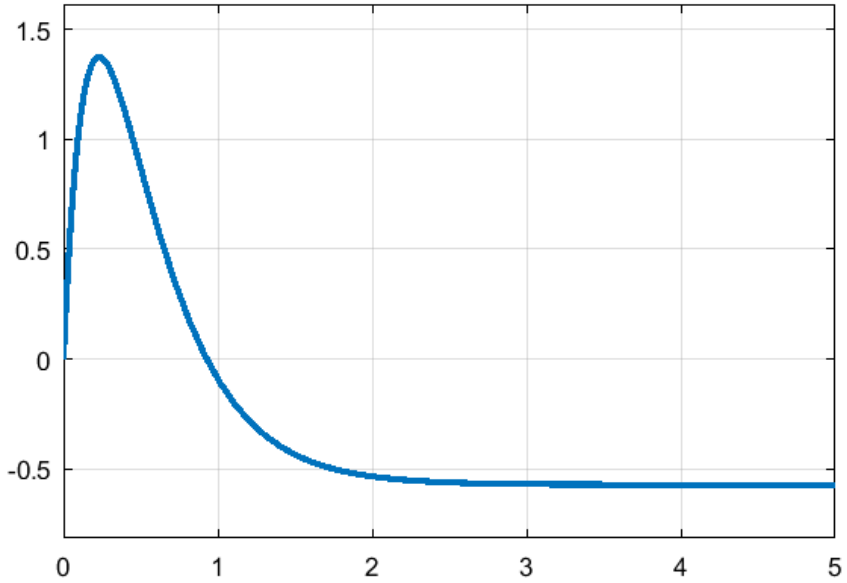


- Unstable, oscillatory



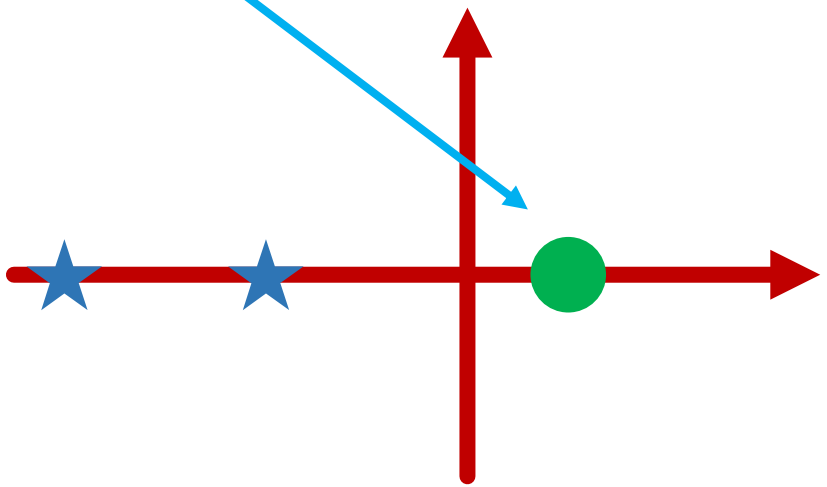
Second Order Example - 3

$$X = \frac{2s-1}{s^2+8s+14}U$$



- Zeros $2s-1=0 \qquad s=0.5$
- Poles $s^2+8s+14=0$

$$s=-4\pm\sqrt{2} \Rightarrow \begin{cases} s\approx -2.6 \\ s\approx -5.4 \end{cases}$$



- Stable but non-minimum phase

「最小相位系统」，其定义为：如果一个稳定的线性时不变系统在复平面右半平面没有零点，则该系统为最小相位系统。

Third Order Example

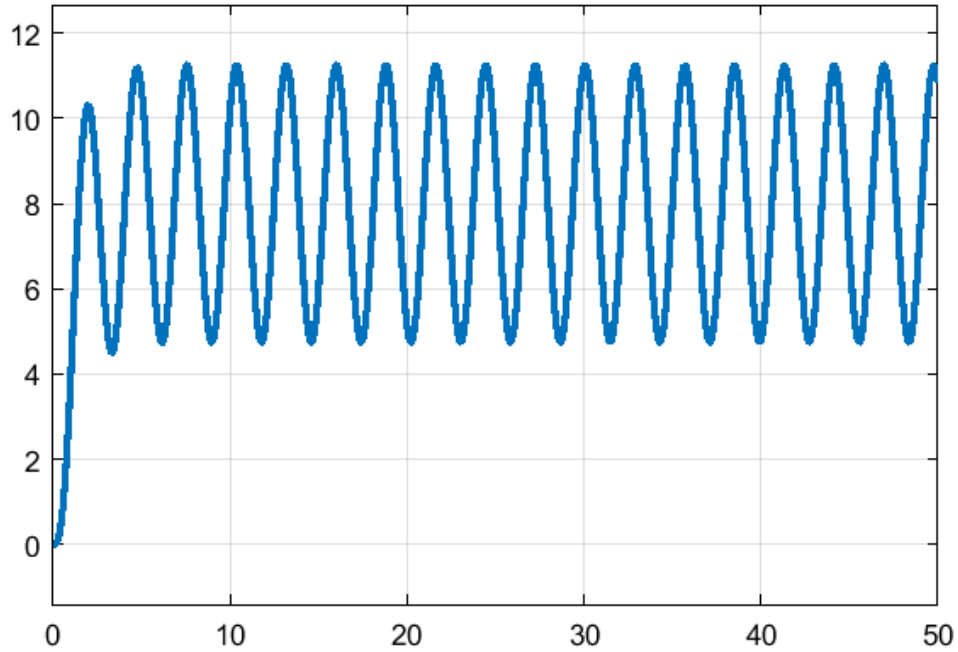
$$X = \frac{5}{s^3 + s^2 + 5s + 5} U$$

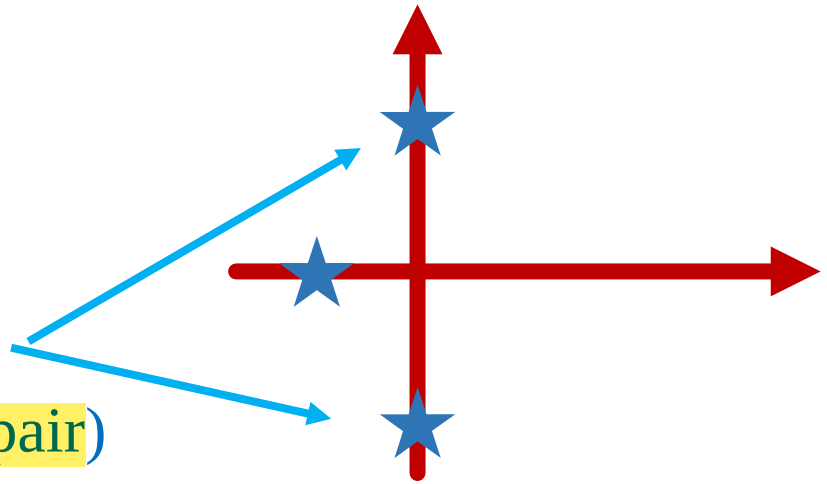
- Zeros None
- Poles $(s+1)(s^2+5) = 0$

(Three solutions, including one quadratic)

$$s = -1 \qquad s = \pm\sqrt{5}\, j$$

- Marginally stable





Dominant poles
(complex conjugate pair)

- Engineering example:
  Hydraulic Piling Rig - 1

$$\frac{dx}{dt} = K \cdot u(t)$$

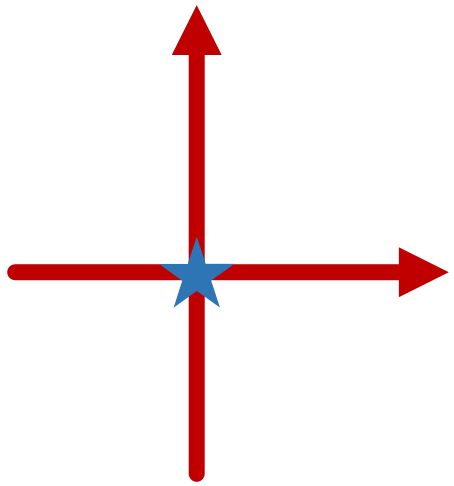


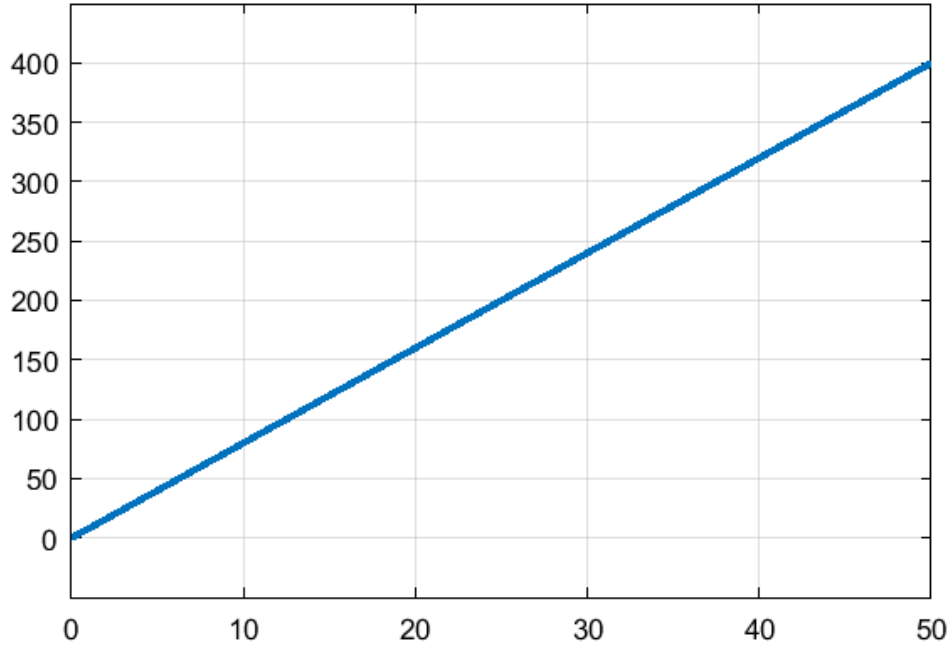
•Engineering example: Hydraulic Piling Rig - 2

- Take Laplace with zero initial conditions

$$\frac{dx}{dt} = K \cdot u(t) \Rightarrow sX = K \cdot U \Rightarrow X = \frac{K}{s}U$$

- Characteristic equation $s = 0$
- Pole at s-plane origin
- Marginally stable





Summary

- Laplace Transforms can be used to solve differential equations
- Zero initial conditions this yields the Transfer Function $G(s) = Z(s)/P(s)$
  - The roots of $Z(s)=0$ (numerator) are called the zeros
  - The roots of $P(s)=0$ (denominator) are called the poles
  - In general these are complex numbers whose position on the Argand Diagram (s-plane) determines the time response
  - The poles define the stability. The system will be asymptotically stable if all the poles have negative real parts